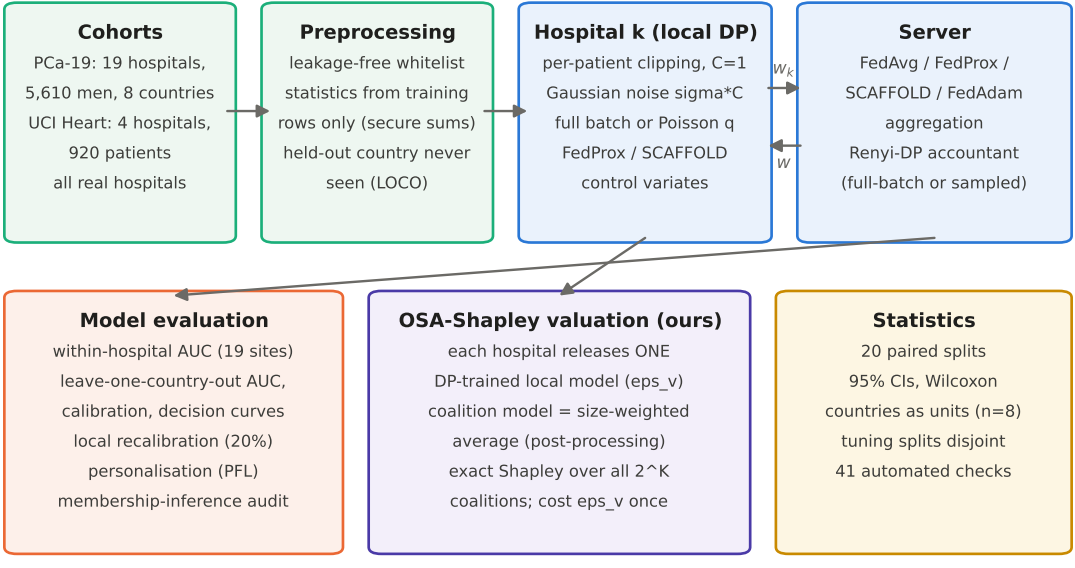
Cohorts
PCa-19: 19 hospitals,
5,610 men, 8 countries
UCI Heart: 4 hospitals,
920 patients
all real hospitals
Preprocessing
leakage-free whitelist
statistics from training
rows only (secure sums)
held-out country never
seen (LOCO)
Hospital k (local DP)
per-patient clipping, C=1
Gaussian noise sigma*C
full batch or Poisson q
FedProx / SCAFFOLD
control variates
w_k
w
Server
FedAvg / FedProx /
SCAFFOLD / FedAdam
aggregation
Renyi-DP accountant
(full-batch or sampled)
Model evaluation
within-hospital AUC (19 sites)
leave-one-country-out AUC,
calibration, decision curves
local recalibration (20%)
personalisation (PFL)
membership-inference audit
OSA-Shapley valuation (ours)
each hospital releases ONE
DP-trained local model (eps_v)
coalition model = size-weighted
average (post-processing)
exact Shapley over all 2^K
coalitions; cost eps_v once
Statistics
20 paired splits
95% CIs, Wilcoxon
countries as units (n=8)
tuning splits disjoint
41 automated checks